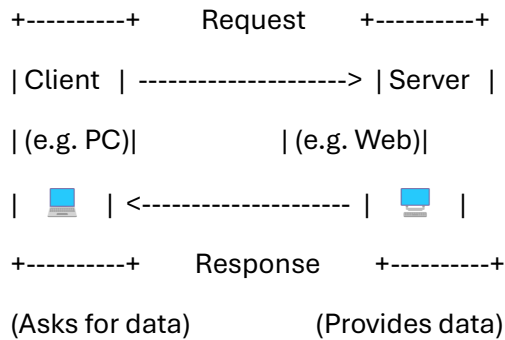


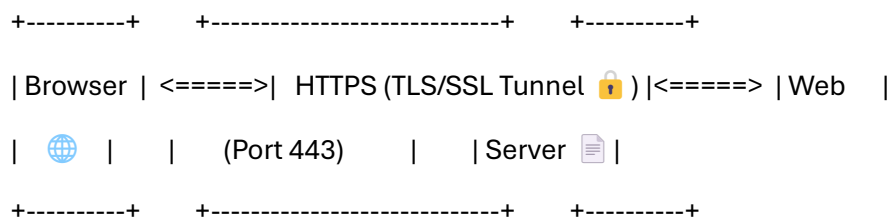
Understanding the Infra: Essential Networking & Security for DevOps

1. Client-Server Architecture:



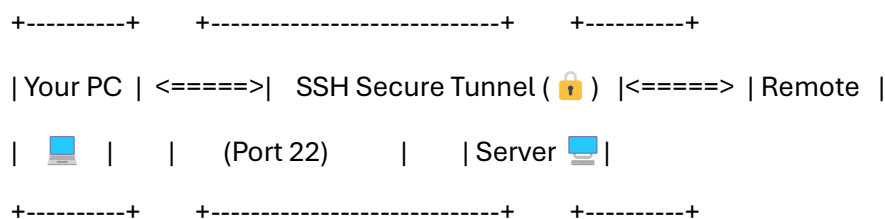
Concept: A client machine requests information or services from a server machine, which processes the request and sends back a response.

2. HTTPS Secure Web Traffic:



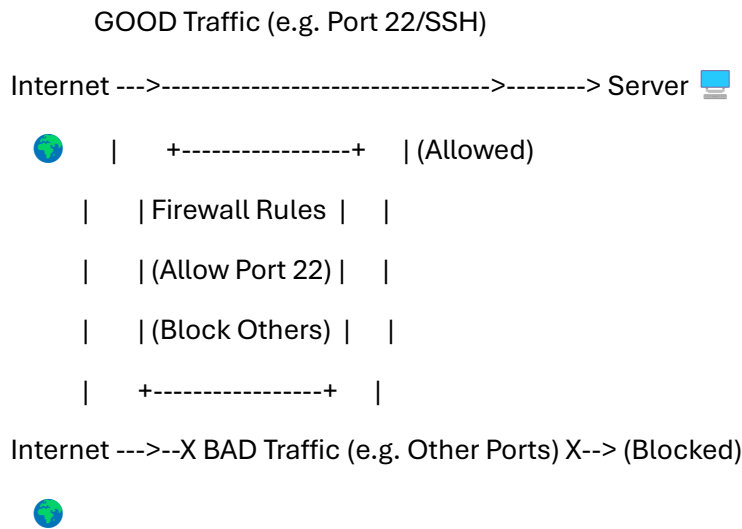
Concept: Your browser communicates with a website through an encrypted tunnel (HTTPS using TLS/SSL) on port 443, protecting the data exchanged.

3. SSH Secure Remote Access:



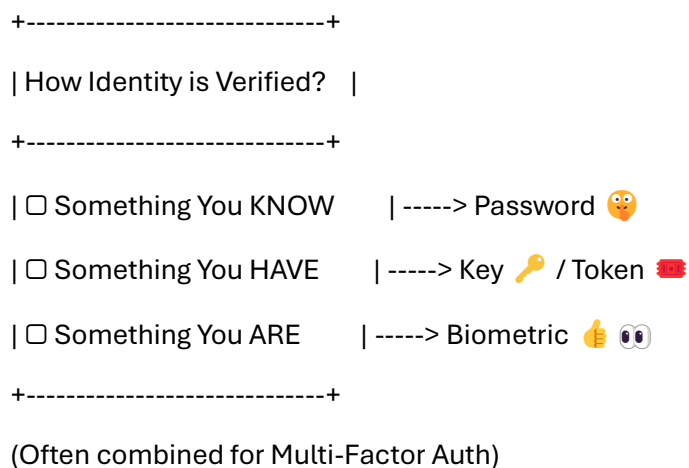
Concept: You connect to a remote server securely over an encrypted SSH tunnel, typically on port 22, allowing command-line access.

4. Firewall Filtering Traffic:



Concept: The firewall inspects traffic, allowing connections on permitted ports (like SSH on 22) while blocking others based on its rule set.

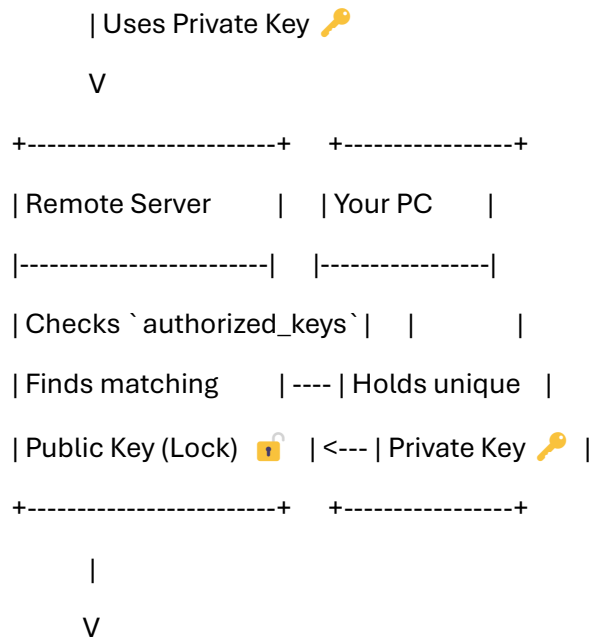
5. Authentication Methods (Conceptual):



Concept: Authentication relies on proving who you are through knowledge, possession, or inherent traits.

6. SSH Key Pair Authentication:

Attempting SSH Login from Your PC...



✅ Access Granted (Keys Match!)

Concept: The server uses the public key (lock) to verify the authenticity of the connection attempt made using the corresponding private key (the only key that fits).

7. SSH Access Setup Flow (Sequence):

[Step 1: Generate Keys ✨ (🔑 + 🔒 on Your PC)] -->

[Step 2: Configure Server Firewall 🛡️ (Allow Port 22)] -->

[Step 3: Upload Public Key 📶 🔒 (Add 🔒 to Server)] -->

[Step 4: Ensure Server is Ready 🟡 (SSH Service Running)] -->

[Step 5: Connect 🚀 (Use 🔑 from Your PC)]

Concept: A sequential process to set up and use secure key-based SSH access.